Activity02

2110413 COMPUTER SECURITY

"Our university is launching a new, all-in-one online portal. This portal will handle everything from student registration and grade management to faculty collaboration and research data storage. The system must protect student privacy, ensure academic integrity, and manage potential conflicts of interest among researchers."

1. Define the security policy for this portal.

The portal has three major security goals:

A. Protect student privacy = Confidentiality [Bell–LaPadula]

Student information such as:

- Personal information
- Grades
- Academic records
- Financial information
- Registration information

must only be accessible to authorized users.

Policy:

- Students can view their own academic records.
- Faculty can view grades of students in their courses.
- Administrative staff can manage registration records.
- Users cannot access student information outside their authorization.

B. Ensure academic integrity = integrity [Biba / Clark–Wilson]

Grades and academic records must not be modified by unauthorized people.

Policy:

- Students can view their grades but cannot modify them.
- Faculty can enter and update grades for their courses.
- Administrative staff can manage official academic records.
- All grade changes should be logged/audited.
- No user should be able to modify grades without appropriate authorization.

C. Protect research data [the least privilege principle]

Research data can contain sensitive information and intellectual property.

Policy:

- Researchers can access research projects they are authorized to work on.
- Researchers can read/write data belonging to their projects.
- Faculty members can manage their own research projects.
- Students/research assistants only receive the minimum permissions necessary.

D. Prevent conflicts of interest [Brewer–Nash (Chinese Wall) model]

Researchers may work with competing research groups or organizations.

Therefore, the portal should enforce conflict-of-interest policies, following the idea of the Brewer–Nash (Chinese Wall) model.

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2. Identify subjects and objects.

Subjects

Subject	Description
Student	Registers for courses and views own grades
Faculty	Teaches courses, manages grades
Researcher	Works with research projects/data
Research Assistant	Assists researchers
Registrar/Admin	Manages student registration and official records

Subject	Description
System Administrator	Manages the portal/system
Research Ethics Officer	Reviews research/conflict-of-interest issues

Objects

Object	Example
Student Profile	Name, student ID, contact information
Registration Record	Courses a student registered for
Grade Record	Exam/course grades
Course Information	Course details
Research Project	Research project metadata
Research Data	Experimental/collected data
Research Documents	Papers, reports, datasets
Faculty Collaboration Data	Collaboration/project information
Audit Log	Record of system activities

3. Choose a primary Access Control Model or Security Framework.

I would choose Role-Based Access Control (RBAC) because a university naturally has **well-defined roles**: Student, Faculty Researcher, Research Assistant, Registrar, System Administrator, Research Ethics Officer

And then **assign permissions to roles**. For example;

- Student: View Own Grades
- Faculty: View + Modify Grades
- Registrar: Manage Academic Records

I can combine it with other security principles:

Requirement	Model / Principle
Student privacy	Bell–LaPadula / Confidentiality
Grade integrity	Biba / Clark–Wilson

Requirement	Model / Principle
Research conflicts	Brewer–Nash
Role permissions	RBAC
Minimum permissions	Least Privilege
Accountability	Auditing

4. Make an Access Control Matrix.

We'll use:

- R = Read
- W = Write

“*” = subject to additional restrictions.

Subject / Object	Student Profile	Registration	Grades	Course Info	Research Project	Research Data	Audit Log
Student	R*	R/W*	R*	R	—	—	—
Faculty	R*	R	R/W*	R/W	R/W*	R/W*	—
Researcher	—	—	—	R	R/W*	R/W*	—
Research Assistant	—	—	—	R	R*	R/W*	—
Registrar/Admin	R/W	R/W	R/W	R/W	—	—	R
System Admin	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Research Ethics Officer	—	—	—	—	R/A	R/A*	R